

FR-018 M1 — Opportunity Discovery Contracts & Architecture

FR018 — sources/optional/m1.md

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M1)** — contracts + authority frozen; executable ingress /
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

Contents

Introduced (career_intelligence.discovery)

Reused (not duplicated)

Deliberately not introduced

Discovery Ingress MAY

Discovery Ingress MUST NOT

Carrier

URL-path minimum (enforced by assert_url_acquisition_provenance)

FR-009 compatibility

FR-018 M1 — Opportunity Discovery Contracts & Architecture

Status: Complete (M1) — contracts + authority frozen; executable ingress / URL adapter deferred to M2

Date: 2026-08-07

Phase: Horizon 1B (lead FR)

Preceding: [M0 spike](#) (**Accepted** — GO under narrow scope)

ADR: [ADR-010](#) (Accepted)

Scope (M1): Production contracts, provenance validators, Protocol surface, documentation. **No** URL fetch, email, feeds, scheduling, Playwright, CLI, or Horizon 1A runtime changes.

Package: `career_intelligence.discovery`

Milestone note: M0's illustrative table listed URL adapter under "M1". Owner M1 brief redefined this milestone as **contracts & architecture only**. Executable URL-first ingress is **M2**.

1. Executive summary

M1 freezes the production design accepted at M0:

```
OpportunitySource → DiscoveryIngress → AcquisitionAdapter
→ ApplicationWorkflowRunner → frozen Horizon 1A pipeline
```

Typed contracts live in `career_intelligence.discovery`. FR-008 `AcquisitionAdapter` / `AcquisitionResult` are **reused**, not redefined. Discovery Ingress is a **Protocol** only — no executable coordinator yet.

Recommendation: Accept M1. Proceed to **M2** (URL adapter + thin ingress implementation + fixtures) only after explicit owner approval.

2. Architecture overview

Layer	Responsibility	M1 deliverable
OpportunitySource	Transient locator	Typed model (url only)
DiscoveryRequest / Outcome	Owner request + per-item results	Typed models
DiscoveryIngress	Thin coordination	Protocol (no impl)
AcquisitionAdapter	One posting acquire	Reuse FR-008
ApplicationWorkflowRunner	Frozen pipeline	Reuse — not called in M1
Opportunity SoT	Durable business record	Reuse FR-009

No second orchestration engine, workflow, or Opportunity store.

3. Production contracts

Introduced (`career_intelligence.discovery`)

Contract	Role
<code>OpportunitySource</code>	Transient <code>source_kind</code> + <code>locator</code> (http/https URL)
<code>DiscoveryRequest</code>	One or more sources; optional <code>force</code> flag (contract only)
<code>DiscoveryItemOutcome</code>	<code>acquired</code> / <code>skipped</code> / <code>failed</code> with consistent fields
<code>DiscoveryOutcome</code>	Aggregate items + count helpers
<code>DiscoveryIngress</code>	<code>discover(request)</code> -> <code>DiscoveryOutcome</code> Protocol
<code>opportunity_source_from_url</code>	Fail-closed URL → source helper
<code>assert_url_acquisition_provenance</code>	Fail-closed URL <code>AcquisitionResult</code> checks

Reused (not duplicated)

Contract	Owner
<code>AcquisitionAdapter</code> / <code>AcquisitionResult</code> / <code>AcquisitionError</code>	FR-008
<code>AcquisitionSourceKind</code> (includes <code>url</code> , ...)	FR-008
<code>OpportunityIdentity</code> / <code>derive_source_facets</code>	FR-009
<code>ApplicationWorkflowRunner</code>	FR-008

Deliberately not introduced

Speculative name	Why omitted
Separate <code>OpportunityProvenance</code> aggregate	<code>AcquisitionResult</code> already carries provenance into <code>AcquireNode</code>
<code>OpportunityAcquisitionMetadata</code> store	Would tempt a second catalogue
Batch <code>acquire_many</code> on adapter	Widens frozen Protocol

4. Authority boundaries

Discovery Ingress MAY

- Resolve `OpportunitySource` values
- Instantiate `AcquisitionAdapter` implementations (M2+)
- Invoke `ApplicationWorkflowRunner.start` (M2+)
- Read Opportunities for definite-identity idempotent **skip** (M2+)
- Return typed `DiscoveryOutcome` items

Discovery Ingress MUST NOT

- Rank opportunities or replace FR-009 ranking
- Assess suitability / run strategy
- Own Opportunity persistence (runner/`PersistOpportunityNode` owns side effects)
- Replace FR-009 duplicate **confirmation** (link/merge policy unchanged)
- Generate documents, submit, advance pipeline, or contact recruiters
- Become a durable "seen jobs" SoT

Recorded in [ADR-010](#).

5. Provenance model

Carrier

`AcquisitionResult` remains the provenance object applied by `AcquireNode`.

URL-path minimum (enforced by `assert_url_acquisition_provenance`)

Field	Requirement
<code>source_kind</code>	Must be <code>"url"</code>
<code>raw_content</code>	Non-empty (model-enforced)
<code>posting</code>	Present
<code>source_url</code>	Required — feeds <code>derive_source_facets</code>
<code>source_identifier</code>	Required — stable locator string
<code>acquired_at</code>	Required
<code>warnings</code>	Optional soft issues

FR-009 compatibility

URL `source_url` enables Seek/LinkedIn/Indeed `platform_job_id` / `canonical_url` derivation.

Fingerprint alone remains non-merge. Idempotent **skip** uses definite identity match →

```
DiscoveryItemOutcome(status="skipped", skip_reason="definite_identity_match",  
matched_opportunity_id=...) .
```

6. Integration with Horizon 1A

Target path (M2+; not executed in M1):

```
DiscoveryIngress.discover  
  → (optional identity pre-check)  
  → AcquisitionAdapter.acquire → assert_url_acquisition_provenance  
  → ApplicationWorkflowRunner.start(adapter)  
  → acquire → ... → persist → owner_review
```

Frozen FR	M1 change
FR-008–017 runtime	None
FR-009 SoT / duplicates	None (contracts cite them)

7. Failure modes (contracted)

Scenario	Representation
Invalid URL	<code>DiscoveryValidationError</code> / source schema reject
Unsupported source kind	M1 allow-list (<code>url</code> only); <code>unsupported_source</code>
Network failure	Item <code>failed</code> + <code>network_failure</code> (M2 adapter)
Adapter failure	<code>adapter_failure</code>
Malformed / empty content	<code>malformed_content</code> / provenance error
Partial metadata	<code>partial_metadata</code> or provenance assert
Definite duplicate	Item <code>skipped</code> + <code>definite_identity_match</code>
Runner failure	<code>runner_failure</code>

All fail closed — no silent success without content/provenance.

8. Testing strategy (M1)

Layer	Coverage
Unit	<code>tests/unit/discovery/test_models_m1.py</code> — source/request/outcome validators, Protocol runtime check, provenance asserts
Contract	Pydantic <code>extra=forbid</code> ; status field consistency
Integration	Deferred to M2 (adapter → runner)
Manual	Deferred to M2 (real owner URLs)
CI	Offline; no network

9. Learning decisions

D1 — Reuse `AcquisitionResult` as provenance

- **Why:** Already applied by `AcquireNode`.
- **Alternative:** New provenance aggregate.
- **Rejected:** Dual models / dual SoT risk.
- **Principle:** Reuse before redesign.
- **Interview:** Extend frozen carriers; don't invent parallel truth.

D2 — Protocol-only Ingress in M1

- **Why:** Freeze authority before fetch complexity.
- **Alternative:** Implement URL adapter in same milestone.
- **Rejected (this M1):** Owner scoped M1 to contracts.
- **Principle:** Validate first; change second.
- **Interview:** Contracts before I/O.

D3 — URL-only `DiscoverySourceKind` in M1

- **Why:** Prevent email/API scope creep in schemas.
- **Alternative:** Allow all `AcquisitionSourceKind` values now.
- **Rejected:** Speculative surface.
- **Principle:** Narrow interfaces.
- **Interview:** Type the milestone you are shipping.

D4 — ADR-010

- **Why:** Ingress authority is a durable decision beyond ADR-003.
 - **Alternative:** Docs-only note.
 - **Rejected:** Academy/acceptance need an ADR handle.
 - **Principle:** Record decisions in the repository.
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10. Documentation updated

Document	Update
This report	Canonical M1
ADR-010	Ingress boundary
adr/README.md	Index
Functional specification § FR-018	M1 status
Domain model	Discovery entities / FR index
Testing strategy	FR-018 M1 section
Implementation notes	M1 sequencing
Roadmap / phase history / changelog	§ 1.117
AGENTS / README / repo guide	Current focus

Historical FR-008–017 acceptance bodies unchanged.

11. Recommendation

Accept M1. Unlock **M2 planning/implementation**:

1. `UrlAcquisitionAdapter` (`source_kind="url"`) with fixture HTTP (no live CI net)
2. Executable thin `DiscoveryIngress` implementation
3. Idempotency pre-check against Opportunity identity
4. Thin owner CLI
5. Manual validation on owner-supplied URLs

Still NO-GO: Playwright, email/feeds, scheduling, scrape-first, Horizon 1A redesign.

12. Final repository status

Item	Status
Horizon 1A	Frozen / unchanged at runtime
FR-018 M0	Accepted
FR-018 M1	Complete
URL adapter / executable ingress / CLI	Not started (M2)
ADR-010	Accepted

M1 COMPLETE

READY FOR M2 (IF OWNER APPROVES)

HORIZON 1A REMAINS FROZEN